



Multiple Choice:

- Find the general solution of the equation $\sin xy' + \cos xy = \ln x$
 - $y = \csc x(x \ln x - x + C)$
 - $y = \cos x(x \ln x - x + C)$
 - $y = \csc x(x \ln x + x + C)$
 - $y = \cos x(x \ln x + x + C)$
- Find the integrating factor of $(y^2 - x)dx + 2ydy = 0$.
 - x^2
 - e^{-x}
 - e^{x^2}
 - e^x
- A population of bacteria grows at a rate proportional to the number present. After two hours, the population has tripled. After two more hours elapsed, the population will have increased by a factor of k . What is the value of k ?
 - 9
 - 6
 - 8
 - 7
- Solve the DE $y' - xy^2 = x$.
 - $y = \tan\left(\frac{x^2}{2} + c\right)$
 - $y = \sin\left(\frac{x^2}{2} + c\right)$
 - $y = \cot\left(\frac{x^2}{2} + c\right)$
 - $y = \cos\left(\frac{x^2}{2} + c\right)$
- Find the orthogonal trajectories of the family of curves $x^2 - y^2 = C^2$.
 - $x^2 + y^2 = C^2$
 - $xy = C$
 - $xy^3 = C$
 - $x - y = C$
- The solution to the DE $\frac{dy}{dt} + 2y = 1$, $y(0) = 1$, is
 - $y = \frac{1}{2}e^{-2t} + \frac{1}{2}$
 - $y = \frac{1}{2}e^{-2t}$
 - $y = \frac{1}{2}e^{-2t} - \frac{1}{2}$
 - $y = \frac{1}{2}e^{-2t} - 1$
- Find the particular solution $y'' - y' - 2y = \sin 2x$.
 - $y_p = -\frac{3}{20}\sin(2x) + \frac{1}{20}\cos(2x)$
 - $y_p = -\frac{3}{2}\sin(x) + \frac{1}{2}\cos(2x)$
 - $y_p = -\frac{1}{20}\sin(2x) + \frac{3}{20}\cos(2x)$
 - $y_p = -\frac{1}{20}\sin(2x) + \frac{3}{20}\cos(3x)$
- The following equation $y^2u_{xx} - 2xyu_{xy} + x^2u_{yy} - \frac{y^2}{x}u_x + \frac{x^2}{y}u_y$ is _____.
 - cyclic
 - hyperbolic
 - elliptic
 - parabolic
- The partial DE $x^2\frac{\partial^2 u}{\partial y^2} + 2xy\frac{\partial^2 u}{\partial x \partial y} + y^2\frac{\partial^2 u}{\partial x^2} = 0$ is an example of _____ equation.
 - heat
 - parabolic
 - hyperbolic
 - elliptic
- Which of the following is a one dimensional heat equation?
 - $\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$
 - $\frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial x^2} = 0$
 - $\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2} + \frac{\partial u}{\partial x}$
 - $\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$
- Given $x^2z - 2yz^2 + xy = 0$. Find $\frac{\partial x}{\partial z}$ at $(1, 1, 1)$.
 - 0
 - 1
 - 1
 - 4/3
- A tank contains 40L of solution containing 2 g of substance per liter. Salt water containing 3 g of this substance per liter runs in at the rate of 4 L/min and the well-stirred mixture runs out at the same rate. Find the time for the amount of substance in the tank becomes 111.1 g.
 - 15 min
 - 37 min
 - 32 min
 - 25 min
- Determine the order and degree of the differential equation $2x\frac{d^4y}{dy^4} + 5x^2\left(\frac{dy}{dx}\right)^3 - xy = 0$.
 - Fourth order, first degree
 - Third order, first degree
 - First order, fourth degree
 - First order, third degree
- Solve the linear equation: $\frac{dy}{dx} + \frac{y}{x} = x^2$.
 - $xy^2 = \frac{x^3}{4} + C$
 - $xy = \frac{x^4}{4} + C$
 - $x^2y = \frac{x^4}{4} + C$
 - $y = \frac{x^3}{4} + C$



15. Which of the following equations is a variable separable DE?

- A. $(x + x^2y) dy = (2x + xy^2) dx$
- B. $(x + y) dx - 2y dy = 0$
- C. $2y dx = (x^2 + 1) dy$
- D. $y^2 dx + (2x - 3y) dy = 0$

16. The population of a country doubles in 50 years. How many years will it be five times as much? Assume that the rate of increase is proportional to the number of inhabitants.

- A. 100 years
- B. 116 years
- C. 120 years
- D. 98 years

17. What is the order of a differential equation?

- A. The order of the highest derivative that appears in the equation.
- B. The order of the lowest derivative that appears in the equation.
- C. The order of the middle derivative that appears in the equation.
- D. The order of the average derivative that appears in the equation.

18. What is the integrating factor for the ordinary differential equation

$$\frac{-2y}{x} dx + (x^2 y \cos y + 1) dy = 0 ?$$

- A. $-2x$
- B. $\frac{-2}{x}$
- C. $\frac{1}{x^2}$
- D. 1

19. Solve $\frac{d^2y}{dx^2} + 5\frac{dy}{dx} + 6y = e^x$.

- A. $C_1 e^{-2x} + C_2 e^{-3x} + \frac{e^x}{12}$
- B. $C_1 e^{-2x} - C_2 e^{-3x} + \frac{e^x}{12}$

- C. $C_1 e^{-2x} + C_2 e^{-3x} - \frac{e^x}{12}$
- D. $C_1 e^{-2x} + C_2 e^{-3x} - \frac{e^{12x}}{12}$

20. Solve the IVP $\frac{dv}{dt} = 9.8 - 0.196v$, $v(0) = 48$.

- A. $v = 50 + 2e^{-0.196t}$
- B. $v = 50 - 2e^{-0.196t}$
- C. $v = 50 + 2e^{0.196t}$
- D. $v = 50 - 2e^{-0.196t}$

21. The solution of the differential

$$\text{equation } \frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 6y = e^{4x} \text{ is } \underline{\hspace{1cm}}.$$

- A. $C_1 e^{2x} - C_2 e^{3x} + \frac{e^{3x}}{3}$
- B. $C_1 e^{2x} + C_2 e^{3x} + \frac{e^{4x}}{2}$
- C. $C_1 e^{2x} + C_2 e^{4x} + \frac{e^{4x}}{4}$
- D. $C_1 e^{2x} - C_2 e^{3x} + \frac{e^{4x}}{2}$

22. The solution of the differential equation $y'' + y = 0$ satisfying the condition $y(0) = 1$, $y(\pi/2) = 2$ is ____.

- A. $y = \cos x + \sin x$
- B. $y = 2(\cos x + \sin x)$
- C. $y = \cos x + 2 \sin x$
- D. $y = 2 \cos x + \sin x$

23. The partial differential equation

$$\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} = 0 \text{ has } \underline{\hspace{1cm}}.$$

- A. degree 1 order 2
- B. degree 2 order 1
- C. degree 1 order 1
- D. degree 2 order 2

24. The differential equation in the form of $y' = P(x)y^2 + Q(x)y + R(x)$ is called a/an ____.

- A. Wave equation
- B. Elliptic equation
- C. Bernoulli equation
- D. Ricatti equation

25. The equation $y' + P(x)y = R(x)y^\alpha$ is ____.

- A. Wave equation
- B. Elliptic equation
- C. Bernoulli equation
- D. Laplacian equation

END